

स्वास्थ्य सारथी

Swasthya Sarathi

AI-Powered Healthcare Triage & District Resource Management

Transforming emergency care in resource-constrained health systems

The Problem

- Manual triage is slow & error-prone
- Healthcare workers overwhelmed at reception
- No real-time visibility of bed/doctor availability
- Medicine stock-outs cause treatment delays
- District officers lack unified oversight
- Critical patients wait too long

- ↑ Mortality rates
- ↑ Patient wait times
- ↑ Staff burnout
- ↓ Resource efficiency
- ↓ Care quality

The Solution

An intelligent, AI-powered platform that:



Instant Triage

Gemini AI analyzes symptoms in seconds



Smart Forecasting

48-hour medicine shortage predictions



Multilingual

English, Hindi, Bengali & Marathi support



Real-Time Visibility

Live bed, doctor, and stock dashboards



Voice-Enabled

Hands-free symptom recording



District Control

Centralized alerts & resource orchestration

End-to-End Workflow

1

Patient — Arrives at clinic

2

Worker — Inputs symptoms (text/voice)

3

Gemini AI — Analyzes & triages (RED/AMBER/GREEN)

4

System — Generates action directive

5

Admin — Monitors resources in real-time

6

District Officer — Receives alerts & coordinates response

Technical Architecture

Frontend Layer

- React (Next.js)
- TypeScript + Tailwind CSS
- Real-time dashboards
- Google Maps integration
- Voice UI for accessibility

AI + Backend

- Google Gemini API
- Firestore Database
- Symptom analysis & triage
- 48-hour forecasting
- Firebase backend
- Cloud-based persistence

AI-Powered Triage Engine

Gemini AI instantly classifies patients:

RED

Critical

Immediate hospital/ICU
admission

Supervised waiting room

General registration & queue

AMBER

Warning

Immediate hospital/ICU
admission

Supervised waiting room

General registration & queue

GREEN

Routine

Immediate hospital/ICU
admission

Supervised waiting room

General registration & queue

Live Resource Monitoring

Grid View

- Beds available per PHC/CHC
- Doctor availability status
- Medicine inventory levels
- Real-time status indicators
- Quick-sort by urgency

Advanced Analytics

- 24-hour historical trends
- Recharts visualizations
- 48-hour shortage forecasts
- Proactive supply alerts
- Consumption rate analysis

Who It Serves



Frontline Healthcare Workers

Register patients, input symptoms, see triage directives



Health Centre Admins

Monitor local resource status, request supplies proactively



District Health Officers (CMOs)

Unified oversight, real-time alerts, resource coordination

Why Now?

- Gemini API is mature & cost-effective
- API can be further migrated to local models
- Firebase enables instant scalability
- React ecosystem is stable & proven
- No proprietary dependencies—fully portable
- Runs on low-bandwidth networks
- Multi-language support built-in
- Cloud-native, minimal infrastructure costs

✓ Ready for piloting in any District Health Department within weeks

Scaling Beyond the Pilot

Phase 1

Pilot (3 months)

Single district • 5-10 health centres • Refine based on feedback

Phase 2

Regional (6-12 months)

Multi-district deployment • 30+ health centres • Advanced analytics

Phase 3

State-Level (18+ months)

1000+ health centres • Mobile app launch • Supply chain integration

Impact Forecast by Year 3

1000+

Health Centres
Connected

500K+

Patients Triageed / Year

40%

Avg. Reduction in Wait
Times

25%

Improvement in Critical
Care Outcomes

Transform Healthcare. Save Lives.

Let's Pilot Swasthya Sarathi in Your District.